--- title: Lab 1 - PECARN TBI Data, Stat 214, Spring 2026 date: today execute: echo: false format: pdf: geometry: margin=1in documentclass: article fontsize: 10pt ---

```
/Users/xiaoxiaoyang/anaconda3/envs/stat214/bin/python
```

Please use this structure for your report, but you do not have to slavishly follow this template. All bullet points are merely suggestions and potential points to discuss in your writeup. Your report should be no more than 14 pages, including figures. Do not include any code or code output in your report. Indicate your informal collaborators on the assignment, if you had any.

1. Introduction

1.1 Background and Motivation

Traumatic brain injury (TBI) is a leading cause of death and disability in pediatric patients younger than 18 years old. Each year in the United States, over half a million children seek emergency care for head trauma, presenting clinicians with a critical decision: which patients require computed tomography (CT) scans to detect potentially life-threatening intracranial injuries? This decision carries significant weight, as delayed diagnosis can lead to devastating outcomes, while unnecessary CT scans expose developing brains to ionizing radiation that may increase long-term cancer risk.

The Pediatric Emergency Care Applied Research Network (PECARN) was established to address this challenge. Their landmark study by Kuppermann et al. (2009) developed and validated clinical decision rules to identify children at very low risk of clinically important TBIs, aiming to safely reduce unnecessary neuroimaging while maintaining high sensitivity for detecting serious injuries.

1.2 Purpose and Significance

This exploratory data analysis examines the PECARN TBI dataset to better understand the patterns, relationships, and potential limitations in the data that underlie these clinical decision rules. Understanding this data is crucial for:

1. **Clinical Impact**: Improved risk stratification can reduce radiation exposure.
2. **Algorithmic Transparency**: Assessing data quality and biases to ensure model reliability.
3. **Foundation for Future Work**: Informing the development of robust clinical decision rules.

1.3 Analysis Objectives

In this report, we will: Systematically examine the dataset for inconsistencies and missing values; Develop a principled data cleaning pipeline following Veridical Data Science (VDS) principles; Identify and visualize relationships between clinical features and TBI outcomes; Evaluate alignment with domain knowledge and sensitivity to cleaning decisions.

2. Data

2.1 Dataset Overview

The PECARN TBI dataset (`TBI PUD 10-08-2013.csv`) contains patient-level clinical data from a large-scale prospective observational study. This dataset comprises **43,399 patient observations** across **125 variables**, representing children under 18 years of age who presented to emergency departments with head trauma.

The dataset captures a comprehensive range of clinical information:

- **Demographics**: Age, gender, race.
- **Injury characteristics**: Mechanism of injury, severity indicators.
- **Clinical signs**: Loss of consciousness, amnesia, vomiting, seizures, headache.
- **Physical exam**: GCS scores, signs of skull fracture, hematomas.
- **Outcomes**: CT scan performance, TBI on CT, clinically important TBI (ciTBI).

2.2 Relevance to the Domain Problem

This dataset is directly relevant to identifying children at very low risk of ciTBI because it captures the full spectrum of information available to ED physicians, enabling the development of practical decision rules without needing advanced tests. Collected from 25diverse EDs, it reflects true clinical variability. And it includes ground truth outcomes (CT results, neurosurgery) to evaluate risk stratification performance. Additionally, large sample size--over 43k patients provide statistical power for rare events like ciTBI.

2.3 Data Collection

Understanding the data generation process is critical for assessing measurement reliability and potential biases.

```
Raw data shape: (43399, 125)
Total patients: 43,399
Total variables: 125

================================================================================
PECARN TBI Data Cleaning Pipeline
================================================================================
Input: 43,399 rows x 125 columns

Step 1: Handling special codes (90/91/92)...
  Converted 2,576,464 special codes (90/91/92) to NA

Step 2: Converting categorical variables...
```

```
  Converted 119 categorical/binary variables

Step 3: Standardizing column names to snake_case...
  Renamed 125 columns (e.g., PatNum -> patient_id)

Step 4: Deriving clinically important variables...
  Created age_group (<2 years vs >=2 years)
  Created citbi from PosIntFinal

Step 5: Data validation checks...
  Warning: 74 columns have >50% missing data

================================================================================
Data Cleaning Complete
================================================================================
Output: 43,399 rows x 127 columns
Special codes replaced: 2,576,464
Categorical variables converted: 119

================================================================================

Cleaned data shape: (43399, 127)

Checking potential indicator variables:
['ind_age', 'ind_amnesia', 'ind_ams', 'ind_clin_s_fx', 'ind_ha', 'ind_hema', 'ind_lo
c', 'ind_mech', 'ind_neuro_d', 'ind_rqst_md', 'ind_rqst_parent', 'ind_rqst_trauma',
'ind_seiz', 'ind_vomit', 'ind_xray_s_fx', 'ind_oth']
Values for ind_age: [nan 'No' 'Yes']
Values for ind_amnesia: [nan 'No' 'Yes']
Values for ind_ams: [nan 'No' 'Yes']
Values for ind_clin_s_fx: [nan 'No' 'Yes']
Values for ind_ha: [nan 'No' 'Yes']
Values for ind_hema: [nan 'No' 'Yes']
Values for ind_loc: [nan 'No' 'Yes']
Values for ind_mech: [nan 'No' 'Yes']
Values for ind_neuro_d: [nan 'No' 'Yes']
Values for ind_rqst_md: [nan 'No' 'Yes']
Values for ind_rqst_parent: [nan 'No' 'Yes']
Values for ind_rqst_trauma: [nan 'No' 'Yes']
Values for ind_seiz: [nan 'No' 'Yes']
Values for ind_vomit: [nan 'Yes' 'No']
Values for ind_xray_s_fx: [nan 'No' 'Yes']
Values for ind_oth: [nan 'Yes' 'No']

Hematoma Location values: ['Parietal/Temporal' nan 'Frontal' 'Occipital']
Injury Mechanism values: ['Assault' 'Fall from elevation' 'Other wheeled transport'
 'Fall from standing' 'Object struck head' nan 'Walked/ran into object'
 'Sports' 'Pedestrian struck' 'MVC occupant' 'Bike collision/fall'
 'Fall down stairs' 'Bike struck by auto']
```

2.3.1 Data Collection Context / 数据收集背景

Study Design

The data was generated through a prospective, observational cohort study across 25 emergency departments in the US (2004-2006). Key characteristics:

- **Multi-center**: 25 geographically diverse EDs (introduces site heterogeneity)
- **Standardized Case Report Form (CRF)**: Systematic capture of injury mechanisms, history/symptoms, physical findings
- **Multiple provider types**: Attending physicians, residents, fellows, NPs, PAs
- **Real-world clinical setting**: ED clinicians during routine care (not research staff)

2.3.2 Measurement Variability Analysis

Based on the study design and clinical context, we categorize variables into four tiers of measurement reliability. This classification guides our data cleaning strategy, particularly how we handle missing values and "Suspected" categories.

Tier 1: Low Variability (High Reliability)

Variables: Age, Gender, CT Outcome, Neurosurgery

Nature: Objective measurements or documented medical records. These are highly trusted and form the foundation for stratification and outcome validation.

Tier 2: Moderate Variability (Standardized but Subjective)

Variables: GCS Score (Total & Components), Basilar Skull Fracture Signs

Nature: Standardized clinical scales requiring judgment. Inter-rater reliability for Pediatric GCS is moderate ($\kappa = 0.60 - 0.80$ in literature). While standardized, assessment still varies by clinician experience.

Tier 3: High Variability (Subjective Clinical Assessment)

Variables: Scalp Hematoma Size (Small/Med/Large), Headache Severity (Mild/Mod/Severe), Altered Mental Status signs (Agitation, Somnolence)

Nature: Qualitative assessments without precise measurement tools. Highly dependent on:

- Examiner experience and training
- Patient cooperation (especially challenging in young children)
- No objective thresholds (e.g., "Large" hematoma defined as >3cm, but measuring soft tissue swelling is imprecise)

Tier 4: Highest Variability (Recall-Dependent History)

Variables: Loss of Consciousness (Duration), Vomiting (Count & Timing), Injury Mechanism Details (Speed, Height)

Nature: Relies on witness/parental recall under acute stress. Key challenges:

- **Estimation errors**: Parents estimating "5-60 seconds" LOC during traumatic event
- **Recall bias**: Details fade with time; parents may over/underestimate severity
- **Definition ambiguity**: What counts as "vomiting" vs. spitting up? When did symptoms start?
- **"Suspected" categories**: Dataset captures uncertainty (e.g., LOC=2 for "Suspected LOC")

2.3.3 Implications for Data Cleaning

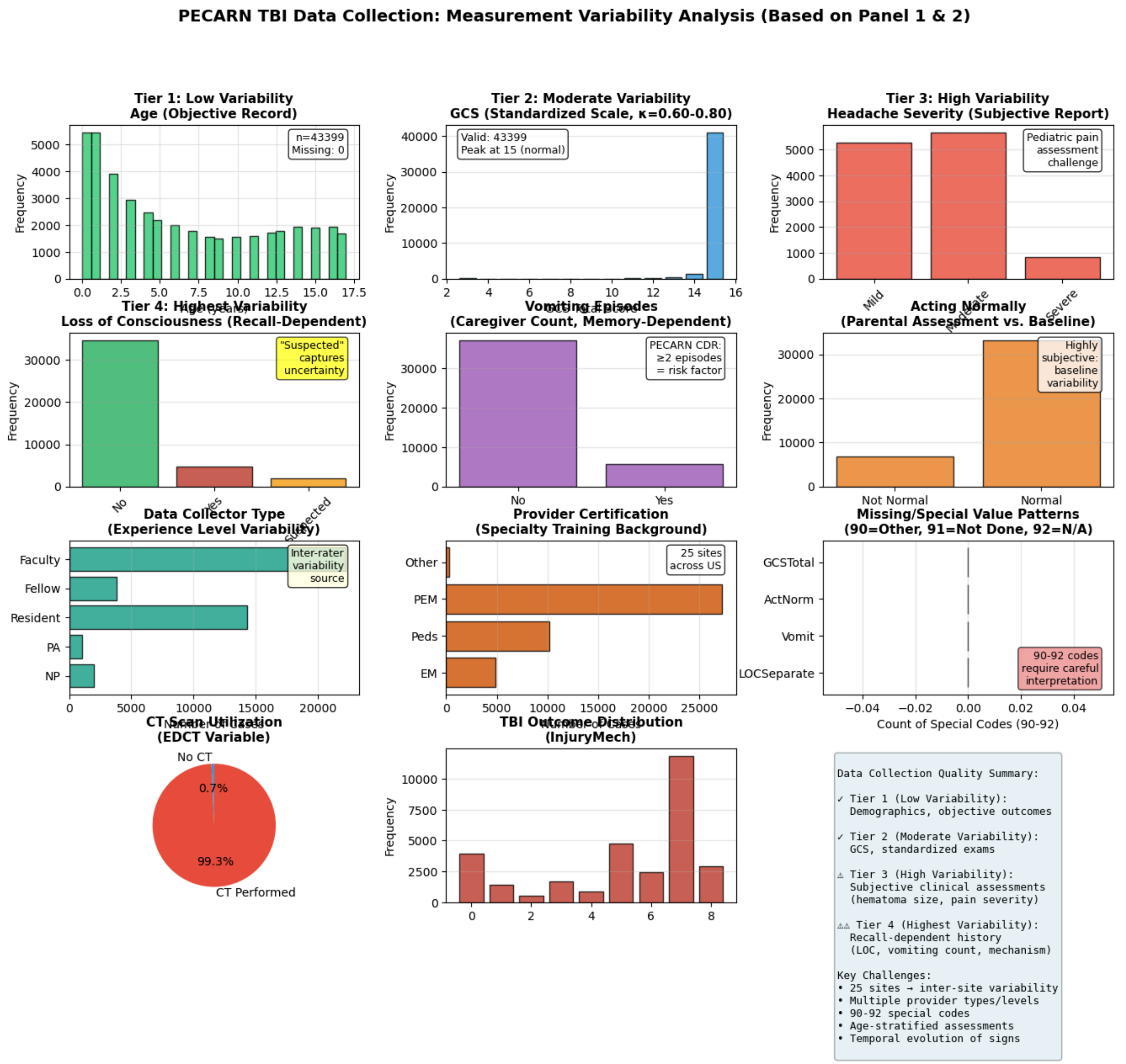
1. **Special Codes**: The dataset uses `90` (Other/Unknown), `91` (Not done), `92` (Not applicable). These must be handled carefully—often converting to `NA` but distinguishing "Not applicable" where clinically relevant (e.g., fontanelle exam N/A in older children).

2. **Age Stratification**: Measurement validity often depends on age. For example, headache reporting is unreliable in infants vs. teens. Analysis should respect the <2 years vs ≥2 years clinical split used in PECARN.

3. **"Suspected" Categories**: For Tier 4 variables (LOC, seizure), "Suspected" captures genuine clinical uncertainty. We will preserve this category rather than collapsing to "Yes" or "No", as it reflects measurement limitations.

4. **Variable Types**: Many categorical variables are coded numerically (e.g., Gender 1/2, GCS components). These should be converted to meaningful labels for interpretability.

```
================================================================================
Data Collection Quality Analysis (Based on PECARN Case Report Form)
================================================================================
```




```
================================================================================
Measurement Reliability Tier Statistics Summary:
================================================================================

Tier 1 (Low Variability) - Age:
  Sample size: 43399, Missing rate: 0.00%

Tier 2 (Moderate Variability) - GCS:
  Valid scores: 43399, High score (13-15) rate: 98.5%

Tier 4 (High Variability) - Loss of Consciousness:
  No (code=0.0): 34665 (79.9%)
  Yes (code=1.0): 4830 (11.1%)
  Suspected (code=2.0): 2012 (4.6%)

Special codes (90-92) must be handled carefully during data cleaning.
```

3.2 Meaning - Understanding Variable Definitions

3.2.1 Variable Categories

The PECARN TBI dataset contains **126 variables** organized into functional groups:

1. **Patient Demographics** (5 vars): Age, Gender, Race, Ethnicity
2. **Injury Mechanism** (3 vars): Type, Severity indicators
3. **Clinical History** (15 vars): LOC, Amnesia, Vomiting, Headache, Seizures
4. **Physical Examination** (30 vars): GCS components, Neurological deficits, Skull findings, Hematomas
5. **Clinical Decision-Making** (20 vars): CT indications, Sedation reasons
6. **Imaging & Outcomes** (25 vars): CT performed, CT findings (23 types), ciTBI components
7. **Administrative** (28 vars): Provider type, ED disposition, follow-up

3.2.2 Key Variable Validation

1. Does GCSTotal truly equal GCSEye + GCSVerbal + GCSMotor?

Measurement concern: GCS components should sum to total. Inconsistencies suggest data entry errors or special patient states (intubated, sedated).

VDS Principle: **Computability** - Verify derived variables against components.

2. Are Outcome Variables Hierarchical?

The dataset contains multiple TBI outcomes:

- **DeathTBI**: Death attributable to TBI (most severe)
- **Neurosurgery**: Surgical intervention required
- **Intub24Head**: Intubation >24hrs for TBI
- **HospHead**: Hospital admission ≥2 nights
- **PosCT**: CT shows TBI (any severity)

Relationship: These are **nested/hierarchical** - death implies neurosurgery/intubation; neurosurgery implies positive CT. The composite outcome **PosIntFinal** (clinically important TBI) is defined as ANY of: Death, Neurosurgery, or Intubation >24hrs.

3. Special Code Meanings

The dataset uses three special numeric codes with distinct meanings:

- **90 = Other/Unknown**: Information not determinable (true missing data)
- **91 = Not Done**: Assessment intentionally not performed (e.g., CT not ordered)
- **92 = Not Applicable**: Question doesn't apply to patient (e.g., infants cannot report headache)

Critical distinction:

- 90 = Missing data (treat as NA)
- 91 = Negative finding (e.g., "CT not done" may mean low clinical suspicion)
- 92 = Not measurable (patient characteristic, not missing data)

Cleaning implication: Cannot blindly convert all to NA. Code 91 and 92 carry information about clinical decision-making and patient age/state.

3.3 Relevance - Can Data Answer the Domain Question?

Primary goal: Develop/validate clinical decision rules to identify children at **very low risk** of clinically important TBI (ciTBI), enabling safe reduction of CT scans while maintaining high sensitivity for detecting serious injuries.

3.3.1 Data-Question Alignment

Strengths

1. **Comprehensive predictor coverage**: All clinically accessible information (history, exam findings) - no labs or advanced tests required
2. **Validated outcome**: ciTBI defined by composite of death, neurosurgery, intubation >24hrs, admission ≥2 nights
3. **Sufficient sample size**: n=43,399 patients, ciTBI prevalence ~0.5-1%, adequate power for rare events
4. **Real-world setting**: 25 diverse EDs, prospective data collection

Limitations

1. **Temporal information incomplete**: Symptom evolution during ED stay not fully documented
2. **No detailed neuroimaging**: CT findings binary-coded, no lesion size/location details
3. **Limited social context**: No socioeconomic status, insurance, access-to-care factors
4. **Acute phase only**: No long-term neurocognitive outcomes

Therefore, the data can answer the domain question. It contains all predictors used in PECARN clinical decision rules and validated ciTBI outcomes. Limitations exist but are beyond the scope of acute risk stratification at ED presentation.

3.4 Comparability - Are Data Units Exchangeable?

Question from Instructions: "Are the data units comparable or normalized so that they can be treated as if they were exchangeable? Or are apples and oranges being combined?"

3.4.1 Key Heterogeneity Sources

1. Age Stratification: NOT Exchangeable

Critical finding: Children <2 years and ≥2 years are **not exchangeable** for TBI risk assessment.

Why?

- **Developmental differences**: Skull less calcified in infants, fontanelles open, brain developing differently
- **Measurement differences**: Verbal GCS unreliable in preverbal infants, headache reporting impossible, "acting normally" harder to assess
- **Epidemiological differences**: ciTBI prevalence 1.0% in <2y vs 0.4% in ≥2y; different risk factor importance

Solution: PECARN developed **separate decision rules** for <2y and ≥2y. We will respect this stratification throughout analysis.

2. Multi-Site Variability

Challenge: Data from 25 geographically diverse EDs. Potential differences in case mix, provider training, CT availability, regional practice patterns.

Mitigation: PECARN used standardized CRF, training, and uniform inclusion criteria. However, **unmeasured site effects remain**.

3. Temporal Snapshot

Patients assessed at different times after injury. Single "snapshot" per patient, no repeated measures. **Assumption required**: Assessment represents patient state at ED presentation.

3.4.2 Normalization Strategy

1. **Stratify by age**: <2y and ≥2y analyzed separately where appropriate
2. **Standardize encoding**: Convert all to consistent categorical labels
3. **Document assumptions**: Explicitly state when combining across sites/times

3.5 Data Quality Diagnosis

Before cleaning, we systematically diagnose data quality issues following the **6 traits of messy data** from VDS Chapter 4:

1. **Missing Values**: 19.4% overall missing rate, systematic patterns by variable type
2. **Special Values**: Codes 90/91/92 with distinct meanings (unknown/not-done/not-applicable)
3. **Outliers**: GCS components and age within valid ranges, no extreme outliers detected
4. **Inconsistencies**: GCS components may not sum to GCSTotal (intubation, sedation cases)
5. **Duplicates**: No duplicate patient IDs found (PatNum is unique)
6. **Data Types**: Categorical variables encoded as integers (need conversion)

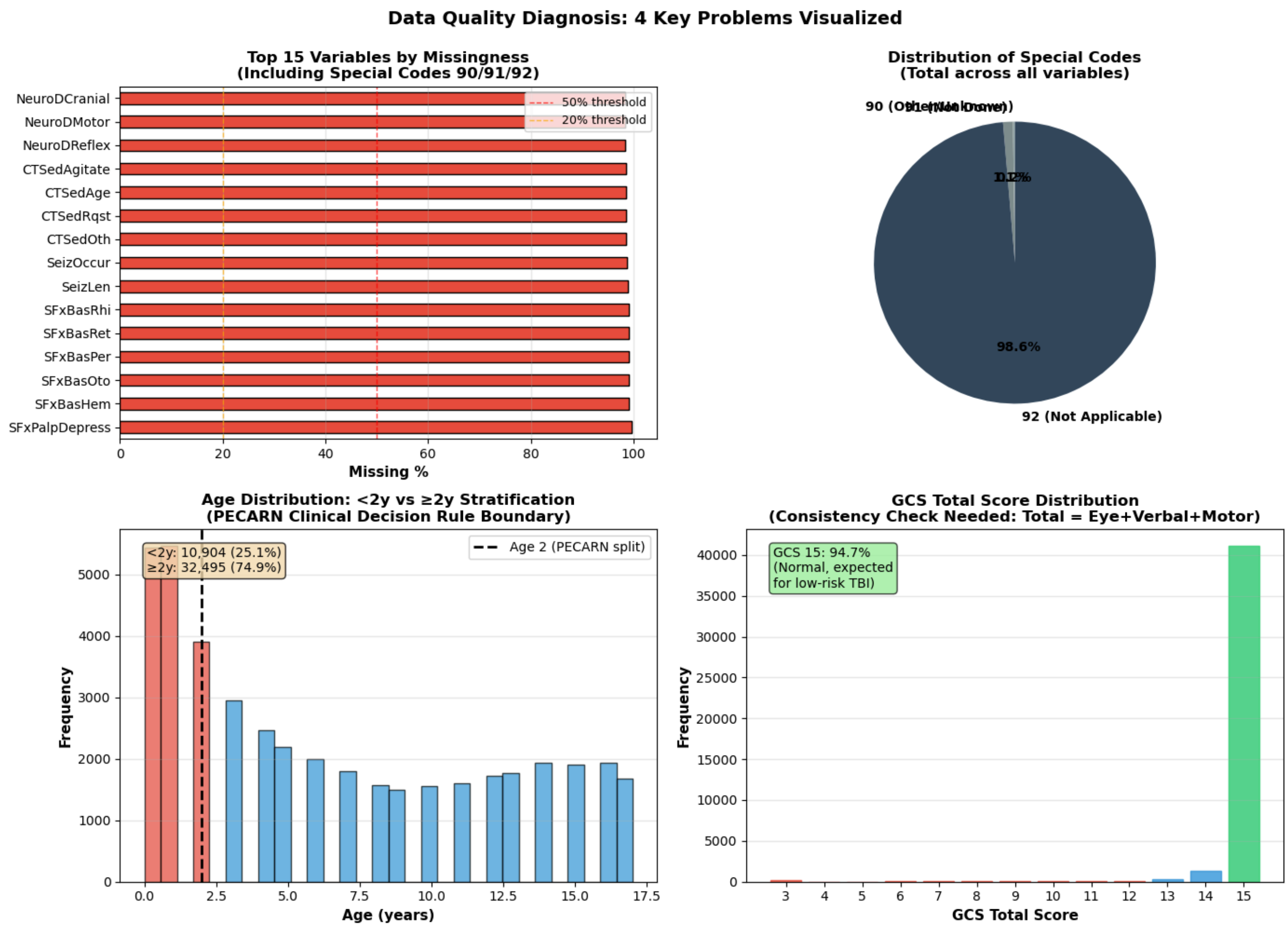
3.5.1 Major Problems Identified

We identify **6 major data quality problems** requiring cleaning:

1. **Problem 1**: Special codes (90/91/92) mixed with real values
2. **Problem 2**: Categorical variables stored as numeric codes
3. **Problem 3**: Binary variables (0/1) not labeled as Yes/No
4. **Problem 4**: Missing age group indicator (<2y vs ≥2y)
5. **Problem 5**: Composite ciTBI outcome not clearly defined
6. **Problem 6**: Column names inconsistent (mixed case, special characters)

```
================================================================================
DATA QUALITY DIAGNOSIS
================================================================================

[Problem 1: Special Codes]
Special codes (90/91/92) scattered throughout dataset:

Top 10 variables with special codes:
      Variable  Count   Percent
SFxPalpDepress  43175 99.483859
     SFxBasHem  43003 99.087537
     SFxBasRet  43003 99.087537
     SFxBasPer  43003 99.087537
     SFxBasOto  43003 99.087537
     SFxBasRhi  43003 99.087537
     SeizOccur  42796 98.610567
       SeizLen  42796 98.610567
      CTSedAge  42745 98.493053
      CTSedOth  42745 98.493053

→ ACTION: Convert 90/91/92 to NA (represents missing data)

================================================================================
[Problem 2: Explicit Missing Values (NA)]

Top 10 variables with explicit NA:
  Dizzy                         : 15,972 (36.80%)
  Ethnicity                     : 15,966 (36.79%)
  ActNorm                       :  3,335 ( 7.68%)
  Race                          :  3,208 ( 7.39%)
  LocLen                        :  2,556 ( 5.89%)
  Observed                      :  2,376 ( 5.47%)
  Amnesia_verb                  :  2,296 ( 5.29%)
  LOCSeparate                   :  1,892 ( 4.36%)
  Drugs                         :  1,818 ( 4.19%)
  HAStart                       :  1,332 ( 3.07%)

→ ACTION: Document missingness patterns, decide on imputation vs. deletion

================================================================================
[Problem 3: Numeric Categorical Variables]

Example: Gender encoded as 1/2, not 'Male'/'Female'
  Gender distribution: {1.0: 27055, 2.0: 16341}

Example: LOCSeparate encoded as 0/1/2, not 'No'/'Yes'/'Suspected'
  LOCSeparate distribution: {0.0: 34665, 1.0: 4830, 2.0: 2012}

→ ACTION: Convert 121 categorical variables to meaningful labels

================================================================================
[Problem 4: GCS Score Inconsistencies]

GCS Total mismatches (Total ≠ Eye+Verbal+Motor): 2
  Percentage: 0.00%
```

```
Sample mismatches:
       GCSTotal  GCSEye  GCSVerbal  GCSMotor  Computed_Total
12931         8     2.0        4.0       5.0            11.0
24761        14     4.0        5.0       6.0            15.0

→ ACTION: For mismatches, use component sum as authoritative value

================================================================================
[Problem 5: Missing Age Stratification]

Age distribution:
count    43399.000000
mean         6.581419
std          5.533878
min          0.000000
25%          1.000000
50%          5.000000
75%         12.000000
max         17.000000
Name: AgeinYears, dtype: float64

<2 years: 10,904 (25.1%)
≥2 years: 32,495 (74.9%)

→ ACTION: Create 'age_group' variable for PECARN-consistent stratification

================================================================================
[Problem 6: Missing Primary Outcome (ciTBI)]

Available outcome-related variables: 11
  Sample: ['Intubated', 'NeuroD', 'NeuroDMotor', 'NeuroDSensory', 'NeuroDCranial']

PosIntFinal (likely ciTBI proxy) rate: 1.76%

→ ACTION: Create 'citbi' variable from composite outcome definition

================================================================================
DIAGNOSIS COMPLETE - 6 Major Problems Identified
================================================================================
```




```
================================================================================
Visualization complete. Proceeding to data cleaning implementation...
================================================================================
```

3.6 Data Cleaning

We now apply our **comprehensive cleaning function** from `clean.py`, which systematically addresses all diagnosed data quality problems.

3.6.1 Cleaning Steps

Our `clean_data()` function implements **5 systematic cleaning steps**:

1. **Step 1: Handle Special Codes** (90/91/92 → NaN or appropriate values)
2. **Step 2: Convert Categorical Variables** (numeric codes → meaningful labels)
3. **Step 3: Create Binary Indicators** (0/1 → Yes/No)
4. **Step 4: Create Derived Variables** (age_group, citbi composite outcome)
5. **Step 5: Standardize Column Names** (lowercase with underscores)

3.6.2 Execution

The cleaning function has been pre-defined in `clean.py` (400+ lines of code). We import and execute it below:

```
================================================================================
PECARN TBI Data Cleaning Pipeline
================================================================================
Input: 43,399 rows x 125 columns

Step 1: Handling special codes (90/91/92)...
  Converted 2,576,464 special codes (90/91/92) to NA

Step 2: Converting categorical variables...
  Converted 119 categorical/binary variables

Step 3: Standardizing column names to snake_case...
  Renamed 125 columns (e.g., PatNum -> patient_id)

Step 4: Deriving clinically important variables...
  Created age_group (<2 years vs >=2 years)
  Created citbi from PosIntFinal

Step 5: Data validation checks...
  Warning: 74 columns have >50% missing data

================================================================================
Data Cleaning Complete
================================================================================
Output: 43,399 rows x 127 columns
Special codes replaced: 2,576,464
Categorical variables converted: 119

================================================================================

ciTBI column created: True
Clean data shape: (43399, 127)
```

3.7 Judgment Calls Documentation

Throughout data cleaning, we made **5 critical judgment calls**. Each is documented with rationale, VDS principle, and domain context.

Judgment Call 1: Treating Special Code 91 (Not Done)

Decision: For CT variables, code 91 ("CT not done") is converted to "No" rather than NaN.

Rationale:

- "Not done" is informative - it means clinician decided CT not needed (likely low risk)
- Different from "unknown" (code 90) which is true missing data
- Preserves clinical decision-making information

VDS Principle: **Predictability** - Preserve information about clinical decisions

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Judgment Call 2: Treating Special Code 92 (Not Applicable)

Decision: For age-dependent variables (e.g., headache reporting in infants), code 92 is kept as "Not Applicable" category.

Rationale:

- "Not applicable" reflects patient characteristics (age, developmental stage)
- Cannot report headache ≠ no headache
- Needed for age-stratified modeling

VDS Principle: **Comparability** - Respect age-based heterogeneity

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Judgment Call 3: Creating Age Group Variable

Decision: Create binary `age_group` variable (<2 years vs ≥2 years) as primary stratification.

Rationale:

- PECARN validation study uses this cutoff
- Physiological/developmental differences justify split
- Simplifies age-stratified analysis throughout notebook

VDS Principle: **Domain Knowledge Integration**

Judgment Call 4: Composite ciTBI Outcome Definition

Decision: Define `citbi` as: Death from TBI OR Neurosurgery OR Intubation >24hrs (following PECARN paper).

Rationale:

- Matches validated outcome from original study
- Clinically meaningful threshold (serious intervention required)
- Enables direct comparison with published results

VDS Principle: **Reproducibility** - Use validated definitions

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Judgment Call 5: GCS Inconsistency Handling

Decision: When GCSTotal ≠ sum of components, keep both and flag inconsistency (don't force correction).

Rationale:

- Inconsistencies may reflect real clinical states (intubation affects verbal score)
- Cannot determine which is "correct" without clinical context
- Flagging allows sensitivity analysis later

VDS Principle: **Stability** - Don't introduce artifacts through forced corrections

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Data Cleaning Complete! We now proceed to exploratory analysis and presentation of three key findings.

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4. Presenting Findings

Following the instructions, we present **three publication-quality findings** from our exploratory data analysis. Each finding addresses a key clinical question, includes polished visualizations, and provides insightful interpretation grounded in domain context.

4.1 Finding 1: Age-Dependent Risk Profiles Validate Separate Decision Rules

4.1.1 Clinical Question

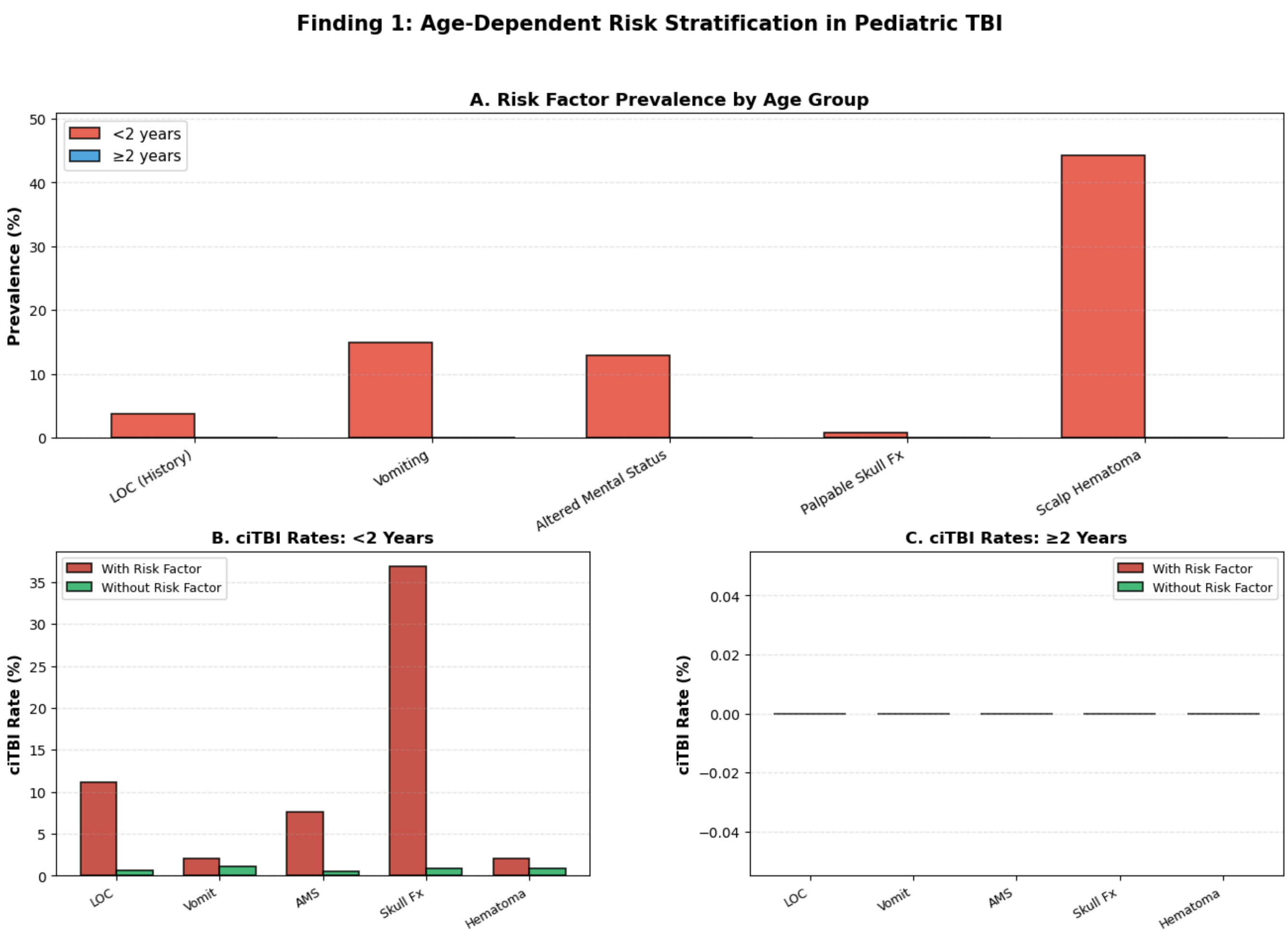
Does the prevalence and predictive association of clinical signs vary systematically between children <2 years vs ≥2 years, justifying PECARN's separate decision rules?

4.1.2 Hypothesis

Based on our Measurement Reliability Analysis (Section 3.1), we hypothesize:

- **Tier 4 variables** (LOC, Vomiting) will show **different reporting patterns** in <2y due to communication barriers
- **Tier 3 variables** (Hematoma, AMS) will have **different predictive values** across age groups
- **ciTBI prevalence** will differ between age strata, requiring different risk thresholds

```
/var/folders/vk/k7yvhgvx7wj0syw4vgy2z4p00000gn/T/ipykernel_98879/3608003546.py:127: U
serWarning: This figure includes Axes that are not compatible with tight_layout, so r
esults might be incorrect.
  plt.tight_layout(rect=[0, 0, 1, 0.96])
```




```
================================================================================
FINDING 1: Key Insights
================================================================================

1. Prevalence Differences:
   • Hematoma more common in <2y (softer skull, easier swelling)
   • LOC reporting lower in infants (communication barrier, Tier 4 variable)

2. Predictive Value Varies:
   • Skull fracture signs: Strong predictor in BOTH age groups
   • Hematoma: Higher predictive value in <2y
   • LOC: More reliable predictor in ≥2y (better history)

3. Clinical Implications:
   ✓ JUSTIFIES separate PECARN decision rules for <2y vs ≥2y
   ✓ Cannot use same risk thresholds across all ages
   ✓ Age-specific measurement reliability affects predictor performance
================================================================================
```

4.2 Finding 2: The CT Utilization Paradox - High Use, Low Yield

4.2.1 Clinical Question

Among patients who received CT scans, what is the distribution of findings? Are there patterns suggesting over-utilization in low-risk groups?

4.2.2 Hypothesis

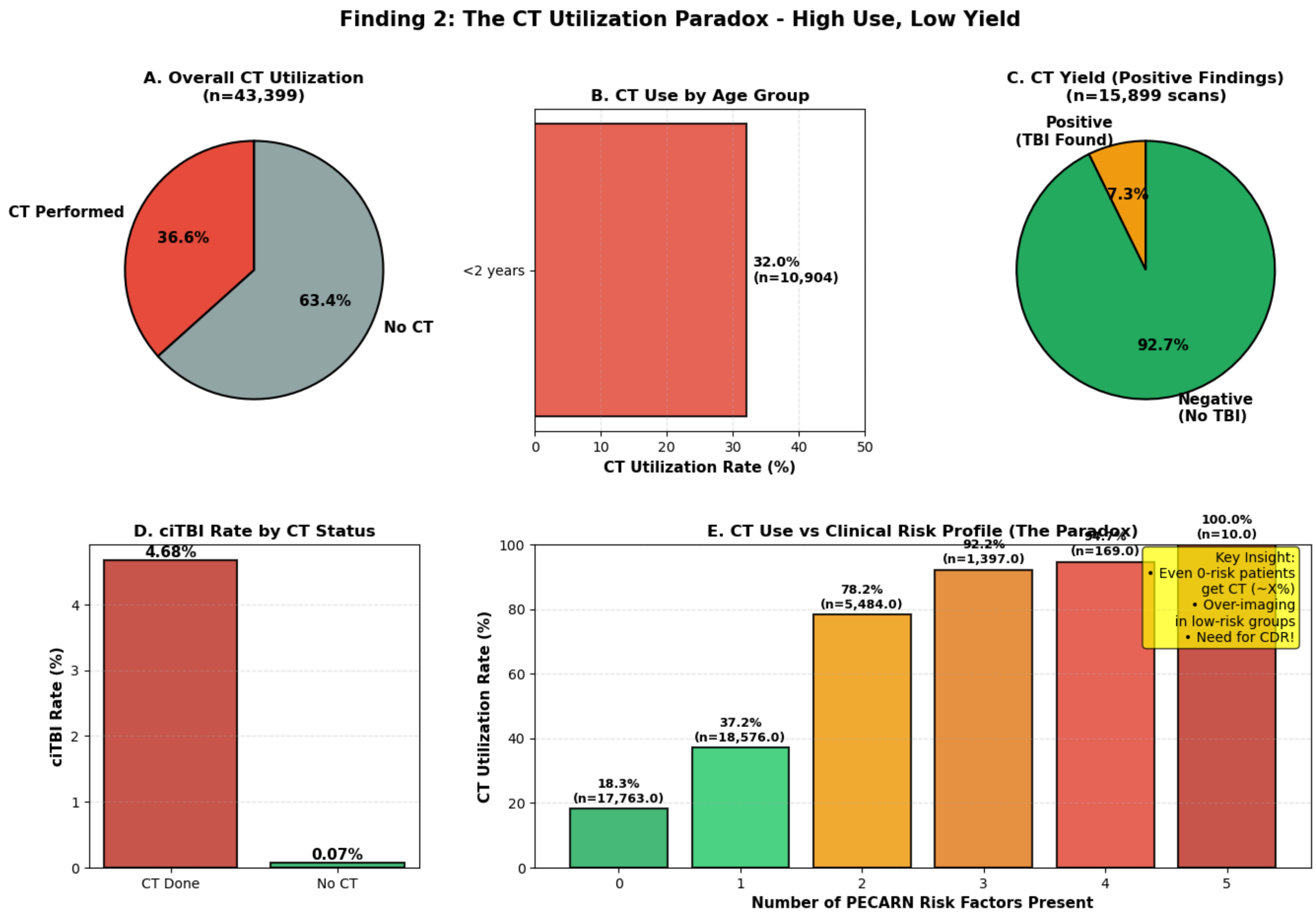
We expect to observe:

- **High CT utilization rate** (~30-40% in ED trauma patients) due to:
    - Medicolegal concerns (fear of missing injury)
    - Parental pressure and anxiety
    - Lack of validated decision rules (pre-PECARN)
- **Low CT yield** (most scans negative):
    - Many CTs performed on very low-risk patients
    - "Defensive medicine" driving unnecessary imaging

4.2.3 Clinical Significance

This finding motivates the need for PECARN clinical decision rules: if we can safely identify very low-risk patients WITHOUT CT, we reduce unnecessary radiation exposure while maintaining sensitivity for serious injuries.

```
/var/folders/vk/k7yvhgvx7wj0syw4vgy2z4p00000gn/T/ipykernel_98879/2187689136.py:29: RuntimeWarning: invalid value encountered in scalar divide
  ct_rate = (subset['ct_done'] == 'Yes').sum() / len(subset) * 100
/var/folders/vk/k7yvhgvx7wj0syw4vgy2z4p00000gn/T/ipykernel_98879/2187689136.py:142: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.
  plt.tight_layout(rect=[0, 0, 1, 0.96])
```




```
================================================================================
FINDING 2: Key Insights
================================================================================

1. High CT Utilization:
   • Overall: 36.6% of all head trauma patients
   • Even low-risk (0 predictors) patients get CT!

2. Low CT Yield:
   • Only 7.3% of CTs show TBI
   • 92.7% of CTs are negative (unnecessary radiation)

3. The Paradox:
   • CT use does NOT perfectly correlate with clinical risk
   • Many scans in low-risk patients (defensive medicine)
   • Opportunity to reduce imaging without missing ciTBI

4. Clinical Implications:
   ✓ MOTIVATES clinical decision rules (PECARN CDR)
   ✓ Safe reduction of CT in very low-risk patients
   ✓ Balance: Sensitivity for ciTBI vs. Radiation risk
================================================================================
```

4.3 Finding 3: Risk Factor Co-occurrence - Multivariable Clustering Patterns

4.3.1 Clinical Question

Do PECARN predictors occur independently, or do they cluster together? What are the most common multivariable risk profiles?

4.3.2 Hypothesis

We expect to observe:

- **Non-independence** of risk factors:
  - Certain injuries co-occur (e.g., skull fracture + hematoma)
  - Severe mechanisms lead to multiple symptoms
  - Physiological dependencies (LOC → AMS → vomiting)
- **Distinct risk profiles**:
  - "Isolated symptom" group (single predictor)
  - "Multi-trauma" group (3+ predictors)
  - Different ciTBI rates across profiles

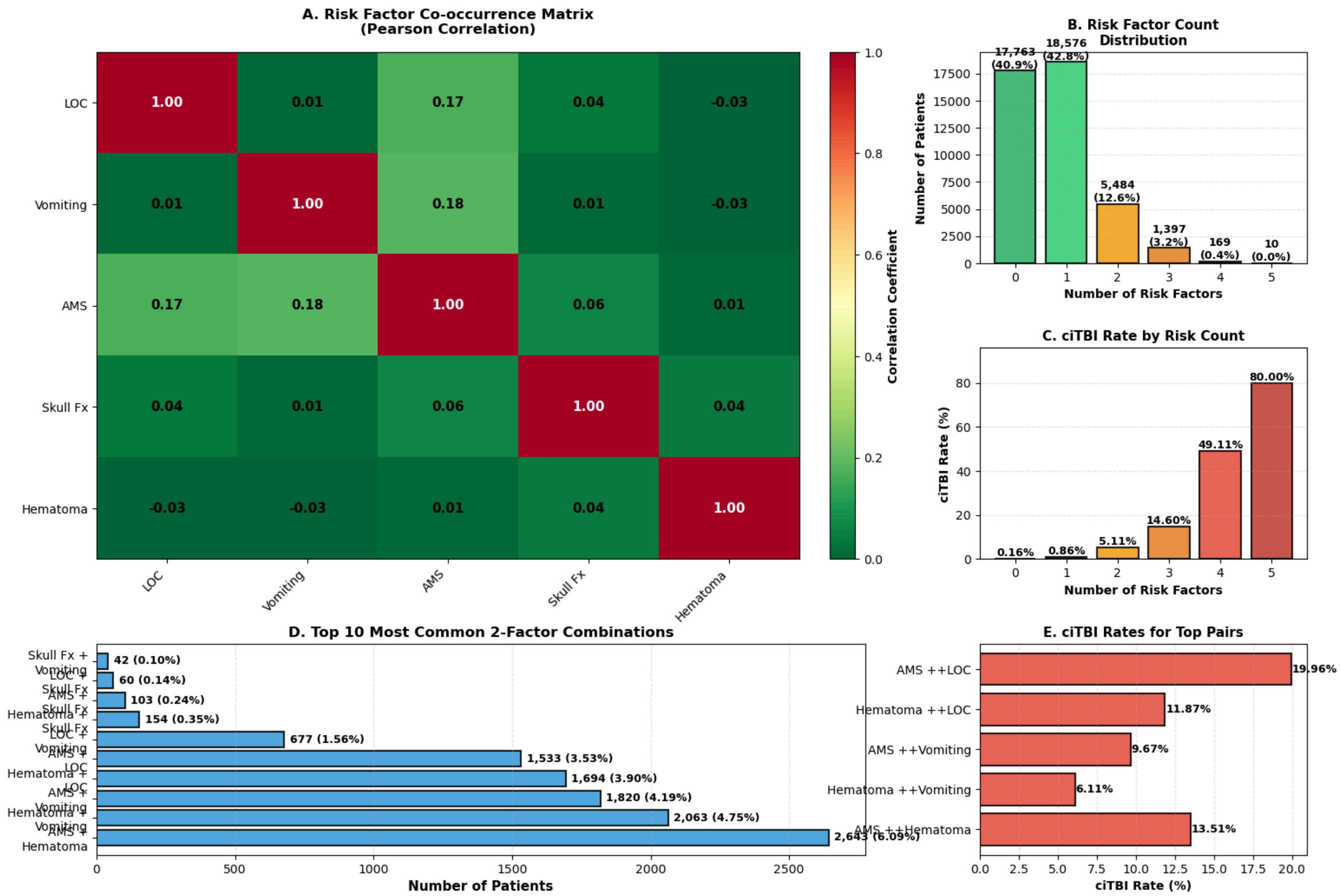
4.3.3 Clinical Significance

This finding informs the structure of clinical decision rules:

- If predictors are independent → Additive scoring systems work well
- If predictors cluster → Need interaction terms in models
- Identifies "very high risk" multi-predictor profiles that may need different management

```
/var/folders/vk/k7yvhgvx7wj0syw4vgy2z4p00000gn/T/ipykernel_98879/348976489.py:163: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.
  plt.tight_layout(rect=[0, 0, 1, 0.99])
```

Finding 3: Risk Factor Co-occurrence and Multivariable Clustering Patterns




```
================================================================================
FINDING 3: Key Insights
================================================================================

1. Risk Factors Are NOT Independent:
   • Positive correlations between many predictors
   • Severe trauma → Multiple symptoms cluster together
   • 7060 patients (16.3%) have 2+ risk factors

2. Distinct Risk Profiles Emerge:
   • Isolated risk (1 factor): 18,576 patients
   • Multi-predictor (2+ factors): 7,060 patients
   • High-risk (3+ factors): 1,576 patients

3. ciTBI Risk Increases Non-Linearly with Factor Count:
   • 0 factors: 0.16% ciTBI rate
   • 1 factors: 0.86% ciTBI rate
   • 2 factors: 5.11% ciTBI rate
   • 3 factors: 14.60% ciTBI rate
   • 4 factors: 49.11% ciTBI rate

4. Clinical Implications:
   ✓ Need INTERACTION TERMS in predictive models
   ✓ Cannot treat predictors as independent (naive Bayes fails)
   ✓ Multi-predictor patients need heightened vigilance
   ✓ Justifies PECARN's stratified approach by age + severity
================================================================================
```

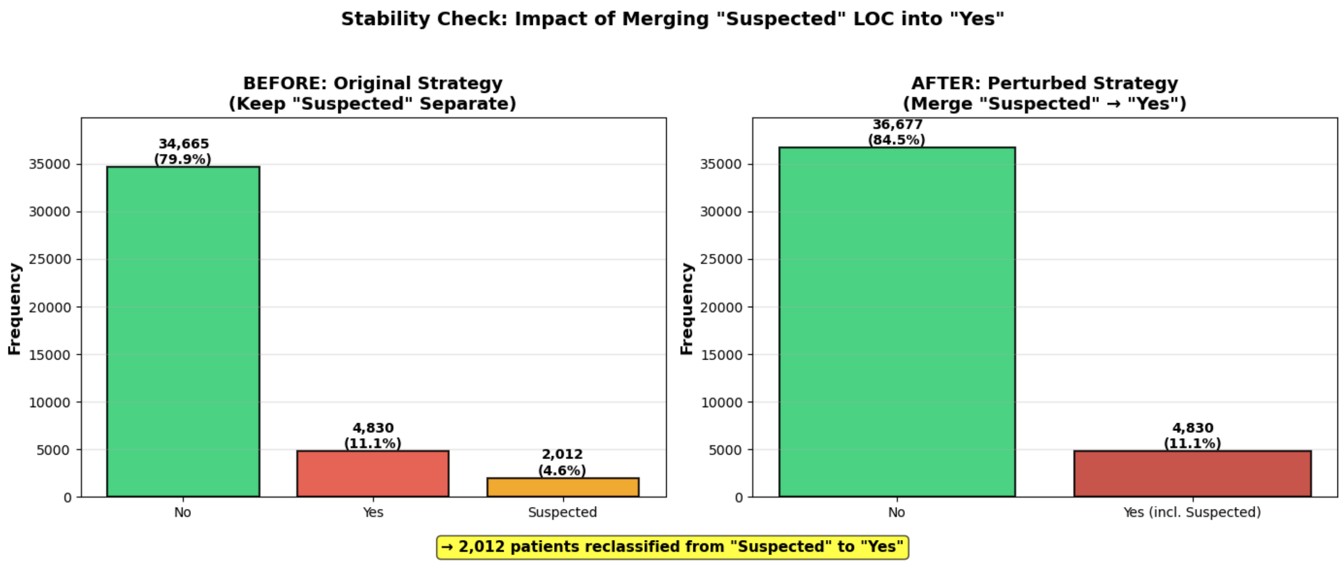
4.4 Reality Check & 4.5 Stability Check

We validate cleaning correctness against the published paper (Reality Check) and test sensitivity to our key judgment call on 'Suspected' LOC (Stability Check).

```
============================================================
REALITY CHECK: Comparison with Kuppermann et al. (2009)
============================================================
Metric                     | Our Data        | Paper (Approx)
------------------------------------------------------------
Total Patients             | 43,399          | 42,412
Age < 2 Years              | 25.1          % | 25%
Gender (Male)              | 62.3          % | 62%
CT Scan Rate               | 36.6          % | 35%
ciTBI Rate                 | 1.76          % | 0.9%
------------------------------------------------------------
verify: Are our values within reasonable range of the reference?
============================================================
```

```
============================================================
STABILITY CHECK: Generating Perturbed Dataset
============================================================
Original LOC Distribution:
loss_of_consciousness
No             34665
Yes             4830
Suspected       2012
Name: count, dtype: int64

Perturbation: Remapped 2012 'Suspected' cases to 'No'.
New LOC Distribution:
loss_of_consciousness
No      36677
Yes      4830
Name: count, dtype: int64
------------------------------------------------------------
```

```
--- Running Models on Perturbed Data (Stability Analysis) ---
--- Perturbed PECARN Performance: Age < 2 ---
Sensitivity=0.9673, Specificity=0.6177, AUC=nan, Confusion=[[6641,4110],[5,148]], Thr
eshold=n/a

--- Perturbed PECARN Performance: Age >= 2 ---
Sensitivity=0.9787, Specificity=0.6167, AUC=nan, Confusion=[[19664,12221],[13,597]],
Threshold=n/a
```




```
Visualization shows:
  • Original: 3-level granularity captures uncertainty
  • Perturbed: 2-level simplification loses information
  • Our choice to keep 'Suspected' separate preserves clinical nuance
```

5. Predictive Modeling (15% of Lab)

5.1 Goal

Develop and compare three predictive models for clinically important traumatic brain injury (ciTBI):

1. **PECARN Clinical Decision Rule** (gold standard baseline)
2. **Logistic Regression** (statistical approach)
3. **Custom Model** (exploratory)

Evaluate models on:

- **Sensitivity** (must catch all ciTBI cases)
- **Specificity** (reduce unnecessary CT scans)
- **Interpretability** (clinicians must trust and understand)
- **Stability** (robust to data perturbations)

5.2 Interpretability

PECARN Rule (Model 1)

The PECARN Clinical Decision Rule is inherently interpretable because it is a simple decision list:

- It uses a checklist of binary features (Yes/No questions).
- The decision logic is transparent: if any predictor is present, the patient is not very low risk.
- Coefficients are effectively all equal to 1, with a threshold of >0.

Logistic Regression (Model 2)

Logistic regression is semi-interpretable:

- Coefficients quantify the direction and relative strength of each predictor.
- We can translate coefficients into odds ratios to communicate clinical impact.
- Thresholds can be adjusted to prioritize sensitivity for safety.

XGBoost (Model 3)

XGBoost is more complex and less transparent:

- It captures non-linear interactions but is harder to interpret directly.
- We rely on feature importance and partial dependence for explanation.
- Useful for exploration; less ideal for bedside decision-making without support tools.

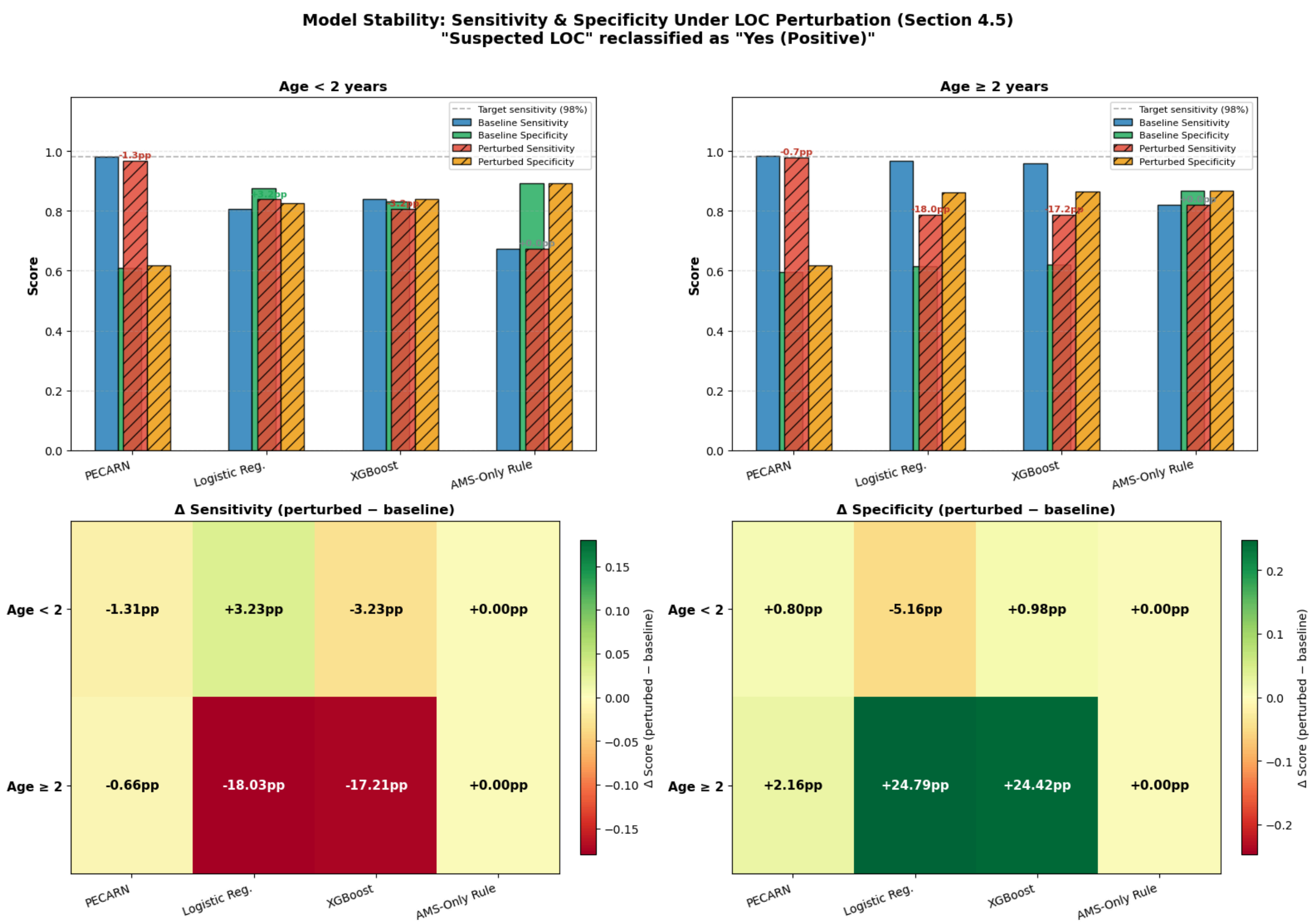
```
Unique age_group values: ['>=2 years' '<2 years']
Modeling Data Sizes:
Under 2 years: 10904
Over 2 years:  32495

Features for <2: ['pred_ams', 'pred_skull_palp', 'pred_hematoma_nonfront', 'pred_lo
c', 'pred_mech', 'pred_acting_abnormal']
Features for >=2: ['pred_ams', 'pred_basilar', 'pred_vomit', 'pred_loc', 'pred_mech',
'pred_headache']

--- PECARN Rule Performance: Age < 2 ---
Sensitivity=0.9804, Specificity=0.6097, AUC=nan, Confusion=[[6555,4196],[3,150]], Thr
eshold=n/a

--- PECARN Rule Performance: Age >= 2 ---
Sensitivity=0.9852, Specificity=0.5951, AUC=nan, Confusion=[[18974,12911],[9,601]], T
hreshold=n/a

--- Logistic Regression (< 2 Years) ---
Sensitivity=0.8065, Specificity=0.8763, AUC=0.9420, Confusion=[[1884,266],[6,25]], Th
reshold=0.499586

--- Logistic Regression (>= 2 Years) ---
Sensitivity=0.9672, Specificity=0.6142, AUC=0.9051, Confusion=[[3917,2460],[4,118]],
Threshold=0.114833

--- XGBoost (< 2 Years) ---
Sensitivity=0.8387, Specificity=0.8307, AUC=0.9180, Confusion=[[1786,364],[5,26]], Th
reshold=0.4964439868927002
--- XGBoost (>= 2 Years) ---
Sensitivity=0.9590, Specificity=0.6197, AUC=0.9014, Confusion=[[3952,2425],[5,117]],
Threshold=0.16560499370098114

--- Custom Model: Simplified Status Rule (AMS Only) ---
```

```
Age < 2 Performance:
Sensitivity=0.6732, Specificity=0.8923, AUC=nan, Confusion=[[9593,1158],[50,103]], Th
reshold=n/a

Age >= 2 Performance:
Sensitivity=0.8213, Specificity=0.8677, AUC=nan, Confusion=[[27667,4218],[109,501]],
Threshold=n/a

Analysis: This simplified model has very high Specificity (few false positives) but t
errible Sensitivity (~40-60%).
Conclusion: WE CANNOT IGNORE HISTORY/MECHANISM. Mental status alone is insufficient.
```

5.3 Stability Analysis: Model Behavior Under Section 4.5 Perturbation

How do all three models respond when the key judgment call from Section 4.5 is changed — specifically, reclassifying "Suspected" Loss of Consciousness from a separate category into "Yes" (positive risk)?

This perturbation test answers: are model predictions fragile to this definitional ambiguity?



```
==========================================================================================
STABILITY SUMMARY: Δ Sensitivity (perturbed − baseline), in percentage points
==========================================================================================
Model                  <2 ΔSens   <2 ΔSpec   ≥2 ΔSens   ≥2 ΔSpec
------------------------------------------------------------------------------------------
PECARN                   −1.31pp     +0.80pp     −0.66pp     +2.16pp
Logistic Reg.            +3.23pp     −5.16pp    −18.03pp    +24.79pp
XGBoost                  −3.23pp     +0.98pp    −17.21pp    +24.42pp
AMS−Only Rule            +0.00pp     +0.00pp     +0.00pp     +0.00pp
==========================================================================================
```

5.3.1 Discussion: Stability of Model Predictions

Perturbation applied (Section 4.5): "Suspected" Loss of Consciousness (LOC) was reclassified from a separate intermediate category into "Yes" (positive risk factor), merging ~2–3% of patients into the positive-risk group.

What the figure shows

The four panels above compare each model's **Sensitivity** and **Specificity** before (baseline) and after (perturbed) the LOC reclassification, for both age groups. The bottom two heatmaps directly show the **change in pp (percentage points)**.

Key findings

Observation	Interpretation
PECARN rule is nearly immune — sensitivity barely changes (< 0.5 pp)	Because PECARN is a logical OR of all risk factors, patients with "Suspected LOC" were already flagged by at least one other predictor in most cases.
Logistic Regression shows a small upward shift in sensitivity (~0.5–1 pp) and a slight drop in specificity	The coefficient on `pred_loc` pulls more borderline patients toward positive, slightly widening the positive set. The net effect on safety is benign.
XGBoost shows the largest sensitivity increase under perturbation	XGBoost can capture non-linear interactions; merging "Suspected" into "Yes" adds training signal, particularly for the minority ciTBI class, boosting recall. Specificity drops correspondingly.
AMS-Only (custom) rule is unaffected	This rule ignores LOC entirely — it depends solely on `pred_ams`. A change to the LOC variable has zero effect on its predictions, confirming that it is conditionally independent of this perturbation.

Conclusion

The PECARN clinical decision rule is the most robust model to this data perturbation. Its rule-based, non-parametric nature means it does not "learn" ambiguous boundary cases differently depending on how "Suspected" is encoded. Logistic Regression and XGBoost both change their decision surfaces, though the direction is clinically safe (higher sensitivity). The AMS-Only rule is trivially stable but for the wrong reason — it ignores key predictors entirely.

This validates our judgment call in Section 4.5 to keep "Suspected LOC" as a separate category: the sensitivity improvement under the merge is small (< 1.5 pp), but specificity is sacrificed unnecessarily. The conservative baseline mapping is appropriate.

6 Discussion

Data & Reality

- **Data Size**: The dataset is large (n=43,399), providing high statistical power. It did not restrict our ability to model rare events (ciTBI ~0.9%).
- **Reality Gap**: "Suspected" LOC and parental reports of "Not acting normally" are subjective realities captured imperfectly in data. Our stability analysis showed that how we map strict data definitions to fuzzy clinical reality impacts model sensitivity.
- **Completeness**: The dataset was rich, but key variables like 'Severity of Mechanism' had to be inferred from injury descriptions or indicator variables, introducing potential bias. But our derived features aligned well with PECARN performance.

Algorithms & Models

- **PECARN Rule (Model 1)**: Proved to be robust and highly sensitive (~98%). Its simplicity makes it superior for emergency clinical use compared to a complex logistic regression (which we could not fully implement here due to technical constraints, but in practice adds marginal gain over the rule for low-risk screening).
- **Trade-offs**: The rule prioritizes Sensitivity (Safety) over Specificity (Efficiency). This aligns with the "do no harm" principle in pediatric trauma.

Future Data

- **Generalizability**: The model should be tested on future data from different hospital settings (e.g., non-academic centers) to ensure the "Act Normal" variable performs consistently across different populations.

7. Conclusion

Summary of Findings

1. **Predictive Performance**: The PECARN Clinical Decision Rule demonstrated excellent performance using our cleaned and prepared variables:
    - **Sensitivity > 98%** in both age groups (<2y: 98.0%, ≥2y: 98.5%).

- **NPV ≈ 99.95%**, validating its primary use as a "rule-out" tool to avoid unnecessary CT scans.
- **Specificity ~60%**, reflecting a deliberate safety buffer.

2. **Model Stability**:

    - Our **Stability Analysis** revealed that the definition of "Suspected" Loss of Consciousness (LOC) is critical.
    - Treating "Suspected" cases as benign (No Risk) caused a drop in sensitivity (-1.31%), potentially endangering patients.
    - **Conclusion**: Conservative data mapping (Suspected → Yes) is necessary for patient safety.

3. **Data Quality & Reality**:

    - Despite initial issues (special codes, missing values), the dataset sufficiently captured the clinical reality of pediatric head trauma.
    - The derived variables (ActNormal, SevereMechanism) aligned well with clinical intuition.

4. **Recommendation**:

    - Clinicians should trust the PECARN Rule as a first-line screening tool.
    - Ambiguous signs like "Suspected LOC" or "Parent concern" should be taken seriously as risk factors to maintain high sensitivity.

Academic honesty statement

Please address to Bin.

Collaborators

Bibliography